MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 4B25

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Full-wave rectifier, xenon gas

See figure 1

Mounting position: Any

Weight: 6 ounces nominal (170.1 grams)

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ip	i surge 0.1 sec	Io	Freq	TA	tk
Unit:	Vac	v	a	aac	Adc	Hz	°C	sec
Maximum:	2.625	700	9.4	360	6.4	150	+70	---
Minimum:	2.375	---	---	---	---	---	-55	40
TEST CONDITIONS:	2.5	700	---	---	---	60	---	40

GENERAL:

Ⓒ Qualification - Not required

Ⓒ denotes changes

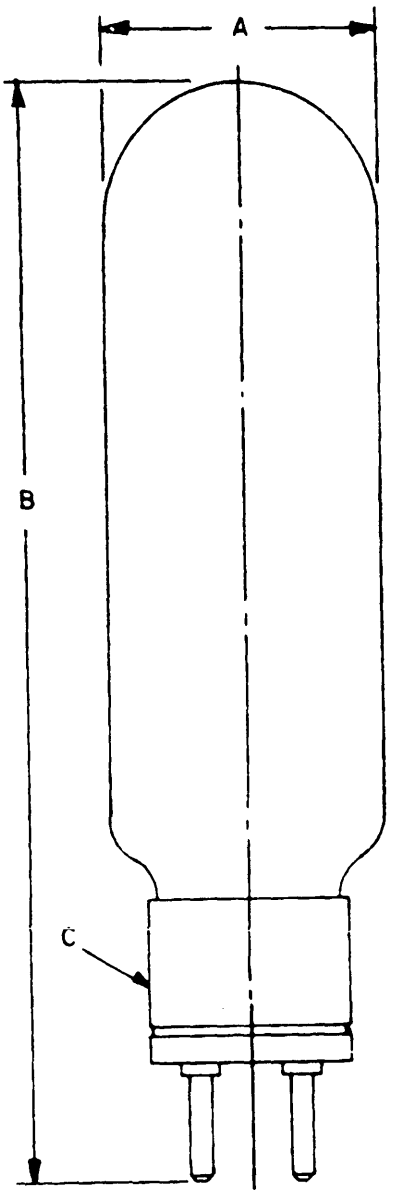
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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MIN	MAX	
	Quality conformance inspection, part 1	1					
1353	Gas rectifiers	2	ib = 9.4 a (min); Io = 6.4 Adc; t = 120 (min)	---	---	---	---
1301	Filament current			If	15.5	18.5	Aac
1256	Electrode current (anode)	3	Ebb = 20 Vdc; Rp = 20 ohms (approx)	Ib	0.1	---	Adc
1231	Peak emission	3,4	ib = 32.0 a	etd	---	20	v
	Quality conformance inspection, part 2						
(C) 1042	Shock, specified pulse			---	---	---	---
	Quality conformance inspection, part 3						
---	Life test		Gas rectifiers; Group D; t = 500 hours	---	---	---	---
---	Life-test end point:						
1231	Peak emission		ib = 32.0 a	etd	---	26	v

NOTES:

1. The AQL for all tests listed in QCI, part 1, shall be 0.65, inspection level II.
2. This test shall be the first test performed at the conclusion of the holding period.
3. Test each unit separately.
4. The inverse current shall not exceed 1 mA during the instant of peak pulse with -700 Vdc applied to the inactive anode.

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PIN CONNECTIONS	
Pin No.	Element
1 & 4	filament
2	anode 2
3	anode 1

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	1.80 (45.7)	2.20 (55.9)
B	7.00 (177.8)	8.50 (215.9)
Quality conformance inspection, part 3 (see note)		
C	Base: A4-53 (EIA)	

NOTE: A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the dimensions shall become a part of QCI, part 2 for three successful consecutive submissions at which time the dimensions may revert to QCI, part 3.

FIGURE 1. Outline drawing of electron tube type 4B25.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3353)

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